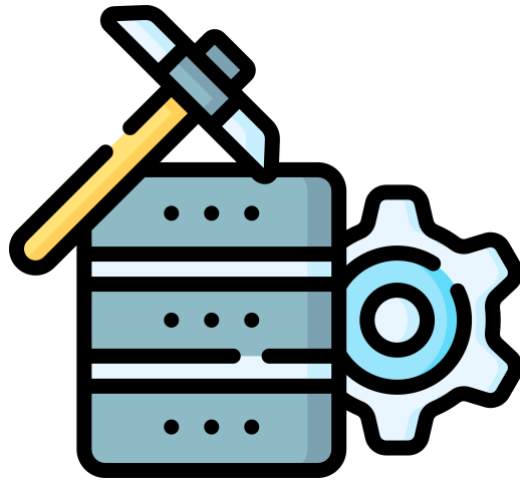




DATA MINING

2025

ORAL & PRACTICAL BANK (Solved)



SENIOR
25

BY QAHWA TEAM

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ORAL BANK

Choose the correct answer:

1.	_____ is an essential process where intelligent methods are applied to extract data patterns .			
	a) Data warehousing	b) Text mining	c) Data mining	d) Data selection
2.	Data mining is best described as the process of _____.			
	a) identifying patterns in data	b) deducing relationships in data	c) representing data	d) simulating trends in data
3.	The various aspects of data mining methodologies is/are _____ I. Mining various and new kinds of knowledge. II. Mining knowledge in multidimensional space. III. Pattern evaluation and pattern or constraint-guided mining. IV. Handling uncertainty, noise, or incompleteness of data.			
	a) I, II, IV only	b) II, III, IV only	c) I, II, III only	d) I, II, III, IV
4.	If the mean < median the data is _____ skewed.			
	a) positively	b) negatively	c) symmetric	d) correlated
5.	If the mean is less than median the data is _____			
	a) Positively skewed	b) Negatively skewed	c) symmetric	d) correlated
6.	If the mean is more than median the data is _____			
	a) Positively skewed	b) Negatively skewed	c) symmetric	d) correlated
7.	If the mean is equal to median the data is _____			
	a) Positively skewed	b) Negatively skewed	c) symmetric	d) correlated
8.	Nonparametric data reduction strategies include all the following <u>except</u> _____			
	a) Histograms	b) Regression	c) Clustering	d) Sampling
9.	Parametric data reduction strategies include _____			
	a) Histograms	b) Regression	c) Clustering	d) Sampling
10.	If you want to <i>give all attributes an equal weight</i> , which preprocess task will you use _____			
	a) Cleaning	b) Integration	c) Transformation	d) Reduction
11.	Task of inferring a model from <u>labeled</u> training data is _____			
	a) Clustering analysis	b) Association rule	c) Transformation	d) Classification
12.	Regression is a method of all the following <u>except</u> _____			
	a) Cleaning	b) Integration	c) Transformation	d) Reduction
13.	The bottleneck of the Apriori algorithm is caused by _____			
	a) No. of rules	b) Min. confidence	c) No. of candidate	d) Correlation
14.	The <u>terminating conditions in decision tree</u> include all the following <u>except</u> _____			
	a) No tuples for a given branch	b) No remaining attributes	c) All tuples belong to the same class	d) No Noise



15.	What is the mean of the following numbers? 10,39,71,42,39,76,38,25 $\text{mean} = \frac{\text{sum of elements}}{\text{number of elements}} = \frac{10 + 39 + 71 + 42 + 39 + 76 + 38 + 25}{8} = 42.5$			
	a) 42	b) 39	c) 42.5	d) 35
16.	The problem that related to calculate the mean is _____			
	a) sensitivity to outlier	b) time consuming	c) noise	d) missing data
17.	If the information gain of age, income, department, and job attributes are 0.24, 0.204, 0.32 and 0.023 respectively which one you will choose as the splitting attribute			
	a) age	b) income	c) department	d) job
18.	If all the proper subsets of an itemset are frequent, then the itemset itself _____			
	a) must also be frequent	b) may be frequent	c) Un frequent	d) is strong
19.	Based on Apriori property, all nonempty subsets of frequent itemset _____.			
	a) must also be frequent	b) may be frequent	c) Un frequent	d) is strong
20.	If the lift measure of the items bread and rice is equal to 1.5 this means bread and rice are _____ (more than 1 → positive)			
	a) positive correlated	b) negative correlated	c) independent	d) uncorrelated
21.	If the lift measure of the items bread and rice is equal to 0 this means bread and rice are _____ (less than 1 → negative)			
	a) positive correlated	b) negative correlated	c) independent	d) uncorrelated
22.	If the lift measure of the items bread and rice is equal to -4 this means bread and rice are _____ (less than 1 → negative)			
	a) positive correlated	b) negative correlated	c) independent	d) uncorrelated
23.	If the lift measure of the items bread and rice is equal to 1 this means bread and rice are _____ (equal to 1 → independent)			
	a) positive correlated	b) negative correlated	c) independent	d) None
24.	The support for the association rule {bread} → (milk) was determined to be 0.95. What does the value 0.95 mean?			
	a) 95% of customers who buy bread buy also milk	b) 95% of customers buy bread and milk	c) 95% of customers who buy milk buy also bread	d) 95% of customers who buy bread don't buy milk
25.	If the value of chi-square test < table value this means the two are _____			
	a) positive correlated	b) negative correlated	c) independent	d) uncorrelated
26.	If the confidence of the rule > the min_confidence this means the rule is _____			
	a) good	b) frequent	c) correlated	d) strong
27.	If the confidence of the rule < the min_confidence this means the rule is _____			
	a) good	b) frequent	c) not strong	d) strong



28.	The full form of KDD is _____			
	a) Knowledge Database Design	b) Knowledge Discovery in Database	c) Knowledge Data Definition	d) Kind Data Distance
29.	The KDD process consists of _____ steps.			
	a) three	b) four	c) five	d) six
30.	The output of KDD is _____.			
	a) Data	b) Query	c) Knowledge	d) Useful information
31.	Various visualization techniques are used in _____ step of KDD			
	a) interpretation	b) data mining	c) transformation	d) selection
32.	This approach is best when we are interested in finding all possible interactions among a set of attributes			
	a) decision tree	b) association rules	c) K-Means algorithm	d) genetic learning
33.	_____ replaces original data volume by smaller data representation .			
	a) Dimensionality reduction	b) Numerosity reduction	c) Compression	d) Transformation
34.	Reducing the number of attributes to solve the high dimensionality problem is called as _____			
	a) Cleaning	b) Overfitting	c) Dimensionality	d) dimensional reduction
35.	This technique uses <u>mean and standard deviation scores</u> to transform real-valued attributes _____			
	a) z-score normalization	b) min-max normalization	c) decimal scaling	d) logarithmic normalization
36.	This is the result of a tuple firing more than one rule with different class predictions _____			
	a) interested rule	b) Strong rule	c) Rule conflict	d) Association rule
37.	Data used to build a data mining model is _____			
	a) validation data	b) training data	c) test data	d) hidden data
38.	The confidence for the association rule $(X) \rightarrow (Y)$ was determined to be 0.90. What does the value 0.90 mean?			
	a) 90% of customers who buy X buy also Y	b) 90% of customers buy Y and X	c) 90% of customers who buy X	d) 90% of customers who buy Y
39.	Correlation analysis is used to _____			
	a) extract rules	b) define support values	c) eliminate misleading rules	d) remove the noise
40.	The <u>correlation</u> coefficient for two real-valued attributes is -0.85 , what does this value tell you?			
	a) As the value of one attribute increases the value of the second attribute also increases			
	b) As the value of one attribute decreases the value of the second attribute increases			
	c) The attributes are not linearly related			
	d) The attributes show a curvilinear relationship			



41.	The <u>correlation</u> coefficient for two real-valued attributes is 0.95 , what does this value tell you?						
	a)	As the value of one attribute increases the value of the second attribute also increases					
	b)	As the value of one attribute decreases the value of the second attribute increases					
	c)	The attributes are not linearly related					
	d)	The attributes show a curvilinear relationship					
42.	Identify the outlier for the given data? 23, 34, 27, <u>7</u> , 30, 26, 28, 31, 34						
	a)	7	b)	23	c)	31	d)
43.	What does the term 'outlier' mean?						
	a)	A score that is left out of the analysis because of missing data					
	b)	The arithmetic mean					
	c)	A type of variable that cannot be quantified					
	d)	An extreme value at either end of a distribution					
44.	Which of the following lists all parts of the <u>five-number summary</u> ?						
	a)	Mean, Median, Mode, Range and Total					
	b)	Minimum, Quartile1, Median, Quartile3 and Maximum					
	c)	Minimum, Maximum, Range, Mean and Median					
	d)	Smallest, Q1, Q2, Q3, Biggest					
45.	Which of the following are <u>interestingness measures</u> for association rules?						
	a)	lift	b)	compactness	c)	recall	d)
46.	Data used to <u>build</u> a data mining model is _____						
	a)	validation data	b)	training data	c)	test data	d)
47.	In the decision tree induction, you will choose the attribute with _____ information gain to be a split						
	a)	large	b)	small	c)	positive	d)
48.	In the decision tree induction, the leaf is _____						
	a)	A question	b)	An attribute value	c)	A class label	d)
49.	In the decision tree induction, the branch is _____						
	a)	A test outcome	b)	An attribute value	c)	A class label	d)
50.	In the decision tree induction, the internal node is _____						
	a)	A test outcome	b)	A test on an attribute	c)	A class label	d)
51.	_____ is the correlation analysis measure for nominal attribute						
	a)	lift	b)	covariance	c)	Correlation coefficient	d)
52.	_____ measure how strongly one attribute implies the other, based on the available data						
	a)	lift	b)	covariance	c)	correlation analysis	d)



53.	_____ is the correlation analysis measure for numeric attributes			
	a) lift	b) covariance	c) Correlation coefficient	d) Chi-square
54.	The SD spread out over a large range of values, this means that the data has_____			
	a) low SD	b) high SD	c) positively skewed	d) negatively skewed
55.	The SD observations tend to be very close to the mean , this means that the data has_____			
	a) low SD	b) high SD	c) positively skewed	d) negatively skewed
56.	For the following association rule: Computer → Webcam (60%, 100%): Which of the following is true? I. 100% of costumers bought both a computer and a webcam II. 60% of costumers bought both a computer <u>and</u> a webcam III. 100% of costumers who bought a computer bought <u>also</u> a webcam IV. 60% of costumers who bought a computer bought also a webcam			
	a) II only	b) III Only	c) I and IV	d) II and III
57.	We have Market Basket data for 1,000 rental transactions at a Video Store. There are four videos for rent -- Video A, Video B, Video C and Video D. The probability that both Video C and Video D are rented at the same time is known as_____			
	a) Correlation	b) support	c) lift	d) confidence
58.	A data_____ represents an entity			
	a) object	b) attribute	c) value	d) none
59.	An_____ is a data field , representing a characteristic or feature of a data object			
	a) object	b) attribute	c) value	d) none
60.	In a binary attribute, _____ both states are equally valuable and have the same weight			
	a) Symmetric	b) Asymmetric	c) Qualitative	d) data
61.	In a binary attribute, _____ states are not equally important			
	a) Symmetric	b) Asymmetric	c) Qualitative	d) data
62.	A finite or <u>countably infinite</u> set of values, integers or otherwise of attribute you will use_____			
	a) interval-scaled	b) ratio-scaled	c) discrete	d) nominal
63.	A measured <u>on a scale of equal-size units</u> of attribute you will use_____			
	a) interval-scaled	b) ratio-scaled	c) discrete	d) nominal
64.	What kind of attribute will you use to define the temperature values?			
	a) interval-scaled	b) ratio-scaled	c) discrete	d) nominal
65.	If a person's height is measured in inches then what kind of attribute you will use_____			
	a) interval-scaled	b) ratio-scaled	c) discrete	d) nominal
66.	The attribute data type of number of telephones in your house is _____.			
	a) interval-scaled	b) ratio-scaled	c) discrete	d) nominal



67.	Bootstrapping allows us to _____.						
	a)	Choose the same training instance several times					
	b)	Build models with alternative subsets of the training data several times					
	c)	Choose the same test set instance several times					
	d)	Test a model with alternative subsets of the test data several times					
68.	_____ fill in missing values, smooth out noise while identifying outliers , and correct inconsistencies in the data						
	a)	Data cleaning	b)	Data integration	c)	Data reduction	d)
69.	Which of the following is the process of detecting and correcting the wrong data ?						
	a)	Data cleaning	b)	Data integration	c)	Data reduction	d)
70.	_____ filling in missing values, smoothing noisy data , identifying or removing outliers , and resolving inconsistencies						
	a)	Data cleaning	b)	Data integration	c)	Data reduction	d)
71.	Removing noise and inconsistent data refers to which phase of the knowledge discovery process?						
	a)	Data cleaning	b)	Data integration	c)	Data reduction	d)
72.	_____ include data from multiple sources in your analysis, map semantic concepts, infer attributes						
	a)	Data cleaning	b)	Data integration	c)	Data reduction	d)
73.	Which of the following is the process of combining data from different sources ?						
	a)	Data cleaning	b)	Data selection	c)	Data integration	d)
74.	_____ obtain a reduced representation of the data set that is much smaller in volume, while producing almost the same analytical results						
	a)	Data cleaning	b)	Data integration	c)	Data reduction	d)
75.	_____ raw data values for attributes are replaced by ranges or higher conceptual levels						
	a)	Data cleaning	b)	Data integration	c)	Data reduction	d)
76.	Concept Hierarchy is used to _____.						
	a)	Data cleaning	b)	Data integration	c)	Data reduction	d)
77.	_____ is the process of <i>finding a model that describes and distinguishes data classes or concepts.</i>						
	a)	Data Classification	b)	Data Characterization	c)	Data integration	d)
78.	_____ is the summarization of the general characteristics or features of a target class of data.						
	a)	Data Classification	b)	Data Characterization	c)	Data integration	d)
79.	_____ replace original data volume by smaller data representation						
	a)	Cleaning	b)	Overfitting	c)	Numerosity reduction	d)
80.	Support and Confidence are measures of _____						
	a)	rule interestingness	b)	association rule	c)	Applications	d)



81.	The measure that are used to determine the <u>degree of extent of variation</u> in a dataset are called ____.			
	a) median	b) mean	c) measures of dispersion	d) measures of central tendency
82.	In this association rule : Computer → antivirus_SW [2%, 60%], Support = _____			
	a) 2%	b) 8%	c) 60%	d) 75%
83.	In this association rule : Computer → antivirus_SW [2%, 60%], Confidence = _____			
	a) 2%	b) 8%	c) 60%	d) 75%
84.	In FP-tree, when you go down the <u>frequency</u> _____.			
	a) increases	b) improves	c) is fixed	d) decreases
85.	Supervised learning differs from unsupervised clustering in that supervised learning requires			
	a) at least one input attribute.	b) input attributes to be categorical	c) at least one output attribute	d) output attributes to be categorical
86.	The correlation between the number of years an employee has worked for a company and the salary of the employee is 0.75. What can be said about employee salary and years worked?			
	a) There is no relationship between salary and years worked.			
	b) Individuals that have worked for the company the longest have higher salaries			
	c) Individuals that have worked for the company the longest have lower salaries			
	d) The majority of employees have been with the company a long time			
87.	Unlike traditional production rules, association rules _____			
	a) allow the same variable to be an input attribute in one rule and an output attribute in another rule			
	b) allow more than one input attribute in a single rule.			
	c) require input attributes to take on numeric values			
	d) require each rule to have exactly one categorical output attribute.			
88.	Given desired class C and population P, lift is defined as _____			
	a) the probability of class C given population P divided by the probability of C given a sample taken from the population.			
	b) the probability of population P given a sample taken from P			
	c) the probability of class C given a sample taken from population P.			
	d) the probability of class C given a sample taken from population P divided by the probability of C within the entire population P			
89.	Association rule support is defined as _____			
	a) the percentage of instances that contain the antecedent conditional items listed in the association rule			
	b) the percentage of instances that contain the consequent conditions listed in the association rule			
	c) the percentage of instances that contain all items listed in the association rule			
	d) the percentage of instances in the database that contain at least one of the antecedent conditional items listed in the association rule			



90.	If the information gain of age, income and gender attributes are 0.42, 0.24 and 0.024 which one will you choose as splitting attribute			
	a) age	b) income	c) gender	d) all of them
91.	Based on Apriori property all nonempty subsets of frequent itemset			
	a) must also be frequent	b) may be frequent	c) can't be frequent	d) all of them
92.	Reducing the number of attributes to solve the high dimensionality problem is called			
	a) cleaning	b) over fitting	c) dimensionality reduction	d) Dimensionality
93.	The bottleneck of the Apriori algorithm is caused by all the following except			
	a) the number of association rules	b) the number of scans required	c) the computations of support for candidates	d) the number of generated candidates
94.	Which of the following is the process of detecting and correcting wrong data			
	a) data cleaning	b) data selection	c) data integration	d) All
95.	If the lift measure of items bread and rice is equal 0.5 this means that			
	a) if client buy bread they are more likely to buy rice	b) if client buy bread they are less likely to buy rice	c) if client buy bread they can buy rice or not with the same probability	d) none of them
96.	The <u>termination condition</u> of the decision tree include the following except			
	a) No tuples for a given branch	b) No noise	c) No remaining attributes	d) All tuples belong to the same class
97.	Correlation analysis is used to _____.			
	a) extract association rules	b) define support and confidence values	c) eliminate misleading rules	d)
98.	If the mean is larger than the median then this might be an indication that the data is _____			
	a) positively skewed	b) symmetric	c) correlated	d) negatively skewed
99.	_____ is the result of tuple firing more than one rule with different class prediction.			
	a) Association rule	b) Strong rule	c) Rule conflict	d)
100.	_____ is the <i>correlation analysis</i> measure for nominal attributes			
	a) Covariance	b) Chi-square	c) Lift	d) Correlation coefficient
101.	Data used to build data mining model			
	a) Validation data	b) Test data	c) Training data	d) Hidden data
102.	Point out the correct statement			
	a) Combining classifiers improves interpretability	b) Combining classifiers reduces accuracy	c) Combining classifiers improves accuracy	d) All of the Mentioned



103.	Point out the correct statement							
	a)	K-means algorithm is used for classification						
	b)	K-nearest neighbor is the same as K-means						
	c)	K-means algorithm aims to partition n objects into k subgroups						
	d)	None						
104.	The K-Means algorithm terminates when _____.							
	a)	A user defined minimum value for the summation of squared error differences between instances and their corresponding cluster center is seen						
	b)	The cluster centers for the current iteration are identical to the cluster centers for the previous iteration.						
	c)	The number of instances in each cluster for the current iteration is identical to the number of instances in each cluster of the previous iteration						
	d)	The number of clusters formed for the current iteration is identical to the number of clusters formed in the previous iteration						
105.	The attribute data type of Number of telephones in your house is							
	a)	Nominal	b)	Ordinal	c)	Interval	d)	Ratio
106.	Which of the following best describes the process of finding the interquartile range for a set of data?							
	a)	ADD the biggest and smallest numbers	b)	Place the number in order from least to greatest then find the middle	c)	Find the difference between the Maximum and the Minimum	d)	Subtract Q3 from Q1
107.	What is the term for the median of the lower half of the data ?							
	a)	Lower Quartile	b)	Upper Quartile	c)	Median	d)	Maximum
108.	What is the term that means the middle data value ?							
	a)	Mean	b)	Median	c)	Mode	d)	Range
109.	What is the mode?							
	a)	# happening the most	b)	the average	c)	biggest - smallest	d)	the middle #
110.	The heights of some students are given.							
	158 cm		172 cm		164 cm			
	164 cm		167 cm		159 cm			
	What is the range of the heights? (range = max – min)							
	a)	13cm	b)	14cm	c)	164cm	d)	330cm
111.	Supervised learning and unsupervised clustering both require at least one _____.							
	a)	hidden attribute	b)	output attribute	c)	input attribute	d)	categorical attribute
112.	Supervised learning differs from unsupervised clustering in that supervised learning requires							
	a)	at least one input attribute	b)	input attributes to be categorical	c)	at least one output attribute	d)	output attributes to be categorical
113.	If a customer is spending more than expected , the customer's intrinsic value is _____ their actual value.							
	a)	greater than	b)	less than	c)	less than or equal to	d)	equal to



114.	Data mining can also applied to other forms such as							
	i) Data streams ii) Sequence data iii) Networked data iv) Text data v) Spatial data							
	a)	i, ii, iii and v only	b)	ii, iii, iv and v only	c)	i, iii, iv and v only	d)	All i, ii, iii, iv and v
115.	Which of the following is not a data mining functionality?							
	a)	Characterization and Discrimination	b)	Classification and regression	c)	Selection and interpretation	d)	Clustering and Analysis
116.	_____ is a <u>summarization</u> of the general <u>characteristics</u> or features of a target class of data.							
	a)	Data Characterization	b)	Data Classification	c)	Data discrimination	d)	Data selection
117.	_____ is a <u>comparison</u> of the general features of the target class data objects against the general features of objects from one or multiple contrasting classes.							
	a)	Data Characterization	b)	Data Classification	c)	Data discrimination	d)	Data selection
118.	Strategic value of data mining is _____							
	a)	cost-sensitive	b)	work-sensitive	c)	time-sensitive	d)	technical-sensitive
119.	_____ is the process of finding a model that describes and distinguishes data classes or concepts							
	a)	Data Characterization	b)	Data Classification	c)	Data discrimination	d)	Data selection
120.	The output of KDD is _____							
	a)	Data	b)	Information	c)	Query	d)	Useful information
121.	Bayesian classifiers is _____							
	a)	A class of learning algorithm that tries to find an <u>optimum classification</u> of a set of examples using the probabilistic theory.						
	b)	Any mechanism employed by a learning system to constrain the search space of a hypothesis.						
	c)	An approach to the design of learning algorithms that is inspired by the fact that when people encounter new situations, they often explain them by reference to familiar experiences, adapting the explanations to fit the new situation						
	d)	None of them						
122.	Classification is _____							
	a)	A subdivision of a set of examples into a number of classes						
	b)	A measure of the accuracy, of the classification of a concept that is given by a certain theory.						
	c)	The task of assigning a classification to a set of examples						
	d)	None of them						
123.	If the mean, median and mode of a distribution are 5, 6, 7 respectively, then the distribution is: (mean < median, 5 < 6, thus it is negatively skewed)							
	a)	skewed negatively	b)	not skewed	c)	symmetrical	d)	bimodal



124.	Which of the following measures of central tendency tends to be most influenced by an extreme score ?			
	a) median	b) mode	c) mean	d)
125.	Which of the following is not a measure of central tendency?			
	a) mean	b) standard deviation	c) median	d) mode
126.	In a group of 12 scores, the largest score is increased by 36 points. What effect will this have on the mean of the scores ? ($36 / 12 = 3$)			
	a) it will be increased by 12 points	b) it will remain unchanged	c) it will be increased by 3 points	d) it will increase by 36 points
127.	Non-parametric data reduction strategies includes all the following except			
	a) Histogram	b) regression	c) clustering	d) sampling
128.	If you want to give all attributes an equal weight which preprocess task you will use			
	a) Cleaning	b) integration	c) transformation	d) reduction
129.	The silhouette coefficient is a method used to determine _____.			
	a) the number of clusters	b) the goodness of clusters	c) the distribution of clusters	d) the density of clusters
130.	All the following are considered as characteristic of the interesting patterns to be discovered through a data mining process except _____.			
	a) no-trivial	b) explicit	c) previously unknown	d) useful
131.	Which of the following clustering requires merging approach ?			
	a) Partitional	b) Hierarchical	c) Density-based	d) Grid-based
132.	In K-nearest neighbor algorithm K stands for _____ while K-means algorithm K stand for _____.			
	a) no. of cluster, no. of neighbors	b) no. of neighbors, no. of clusters	c) no. of rules, no. of classes	d) no. of classes, no. of rules
133.	_____ algorithm is bottom-up composition and start from each object has its own cluster.			
	a) K-mean	b) Agglomerative	c) Divisive	d) B & C



True/False:

1.	Association rules provide information in the form of "if-then" statements. (Decision tree)	F
2.	<u>Binning</u> is a method of reduction . (cleaning)	F
3.	Cluster is the process of finding a model that <u>describes and distinguishes data classes or concepts</u> . (Classification)	F
4.	<u>False Negatives</u> means <u>positive</u> tuples <u>incorrectly</u> labeled.	T
5.	<u>Intrinsic methods</u> measure how well the clusters are separated	T
6.	Mode can be found in the <u>nominal attribute</u> values.	T
7.	Recall is interestingness measures for association rules. (Lift)	F
8.	<u>Sorting a student database</u> based on student identification numbers is a data mining task. (database query)	F
9.	The goal of clustering analysis is to maximize the number of clusters . (maximize the intra-cluster similarity and minimize the inter-cluster similarity)	F
10.	The probability that two items are <u>bought together</u> is known as confidence . (support)	F
11.	The silhouette coefficient is a method to determine the natural number of clusters for hierarchical algorithms density-based algorithm (partitioning algorithms)	F
12.	All continuous variables are ratio. (Some)	F
13.	<u>Attributes</u> are sometimes called <u>variables</u> and <u>objects</u> are sometimes called <u>observations</u>	T
14.	Binary variables are sometimes continuous (Numeric with two values)	F
15.	Computing the <i>total sales of a company</i> . Is a data mining task (accounting)	F
16.	Correlation analysis divides data into groups that are meaningful, useful, or both. (Cluster analysis)	F
17.	Database mining refers to the process of deriving high-quality information from text (Text mining)	F
18.	Dissimilarity matrix stores n data objects that have p attributes as an n-by-p matrix (Data)	F
19.	Dividing the customers of a company according to their profitability. Is a data mining task (accounting)	F
20.	For an association rule, if we move one item from the right-hand-side to the left-hand-side of the rule, then the confidence will never change. (support)	F
21.	If all the <u>proper subsets</u> of an itemset are frequent, then the itemset itself <u>must also be frequent</u> . (may)	F
22.	In decision tree algorithms, <u>attribute selection measures</u> are used to <u>rank attributes</u>	T
23.	In decision tree algorithms, attribute selection measures are used to <u>reduce the dimensionality</u> .	T
24.	In lazy learner we interest in the largest distance (minimum)	F
25.	<u>Intrinsic methods</u> measure <u>how well the clusters are separated</u>	T
26.	Multimedia Mining is the application of data mining techniques to discover patterns <u>from the Web</u> . (Web data mining)	F
27.	Regression is a method of integration (correlation)	F
28.	Strategies for data transformation include chi-square test (integration)	F
29.	The Pruning make the decision tree more complex (reliable)	F
30.	An object is an outlier if its density is equal to the density of its neighbors. (significantly lower than)	F
31.	A common weakness of association rule mining is that it is not produce enough interesting rules (it produces too many rules)	F
32.	Accuracy is interestingness measures for association rules (correlation, lift)	F
33.	Core object is an object whose ϵ -neighbourhood contains objects less than MinPts (more than)	F
34.	<u>Correlation analysis</u> is used to <u>eliminate misleading rules</u>	T
35.	Correlation is a method of cleaning (analysis)	F



36.	Data matrix stores a collection of proximities for all pairs of n objects as an <u>n-by-n</u> matrix (Dissimilarity)	F
37.	Extracting the frequencies of a sound wave. Is a data mining task. (signal processing)	F
38.	Incomplete data problem can be solved by <i>binning</i> (binning is a smoothing technique)	F
39.	K-Nearest Neighbor <u>Classifiers</u> <u>do classification</u> when <u>new test data is available</u>	T
40.	Median is a value that occurs most frequently in the attribute values (Mode)	F
41.	Mode is a middle value in set of ordered values (Median)	F
42.	One strength of a <u>Bayesian Classifier</u> is that it <u>can be easily trained</u>	T
43.	Outlier analysis is a method of transformation (cleaning)	F
44.	Predicting the outcomes of tossing a (<u>fair</u>) pair of dice. Is a data mining Task? (probabilistic) (If the die were not fair this could be data mining task)	F
45.	Redundancy is an important issue in data cleaning (integration)	F
46.	Sampling methods smooth noisy data (reduction)	F
47.	The bottleneck of the Apriori algorithm caused by the number of association rules . (number of candidates and scans)	F
48.	the object is local outlier if it is deviate significantly from the rest of the dataset (global)	F



WRITTEN BANK

Answer the following questions:

1. Define: A centroid in k-means.

Answer: A centroid in K-Means is the central point of a cluster, calculated as the mean of the coordinates of all points assigned to that cluster. It is updated iteratively to minimize the distance between points and the cluster center.

2. Define: A core point in DBSCAN.

Answer: A core point in DBSCAN is a point that has at least MinPts points (including itself) within its ϵ -neighborhood, forming the core of a cluster.

3. Define: Association and correlation analysis. Give an example.

Answer:

Association analysis identifies frequent itemsets and generates rules (e.g., $\{A\} \rightarrow \{B\}$) using measures like support and confidence

Correlation analysis measures the strength and direction of relationships between variables (e.g., correlation coefficient). Example: Association rule $\{\text{Milk}\} \rightarrow \{\text{Bread}\}$ with 10% support and 80% confidence; correlation of 0.7 between milk and bread sales indicates a strong positive relationship.

4. Define: Cluster analysis. Give an example.

Answer: Cluster analysis groups similar data points into clusters, maximizing intra-cluster similarity and minimizing inter-cluster similarity, using unsupervised learning. Example: Grouping customers by purchasing behavior (e.g., health-conscious vs. fast-food buyers).

5. Define: Data Cleaning, Data Integration, Data Reduction, Data Transformation, Discretization.

Answer:

1-Data Cleaning: Removing noise, handling missing values, and correcting errors to improve data quality.

2-Data Integration: Combining data from multiple sources into a unified view, resolving redundancies.

3-Data Reduction: Reducing data volume while preserving key information (e.g., dimensionality reduction).

4-Data Transformation: Converting data into a suitable format (e.g., normalization).

5- Discretization: Converting continuous data into discrete categories (e.g., age to "young," "adult").



6. Define: Outlier analysis. Give an example.

Answer: Outlier analysis identifies data points that deviate significantly from the rest.
Example: In student grades (60-90 range), a **grade of 10 is an outlier**.

7. Define: Regression. Give an example.

Answer: Regression predicts a continuous dependent variable based on independent variables ---> Example: Predicting store sales based on advertising budget.

8. Give an example of nonparametric data reduction strategies.

Answer: Histogram-Based Reduction: Dividing data into bins and replacing values with bin representatives

9. Give an example of parametric data reduction strategies.

Answer: Regression-Based Reduction: Using a linear regression model to represent data with a simpler equation, reducing dimensionality.

10. How K-Means differs from DBSCAN?

Answer: K-Means partitions data into k predefined clusters, assuming spherical shapes, using centroids. DBSCAN forms clusters based on density, doesn't require specifying k, detects arbitrary shapes, and identifies outliers as noise.

11. How to assess the goodness of a rule?

Answer: The goodness of a rule is evaluated by:

1- Support: Percentage of transactions containing the rule's items.

2-Confidence: Percentage of transactions with the antecedent also containing the consequent.

3-Lift: Strength of the rule compared to independence (>1 is positive).

4-Conviction/Leverage: Additional measures of rule strength. A good rule has high support, confidence, and lift > 1 .

12. How can we handle missing values?

Answer:

Ignore tuples with missing values

Fill with a global constant (e.g., "Unknown") attribute mean/median

Predict using methods like decision trees or Bayesian inference

Manual filling for small datasets.

13- If a person's height is measured in inches, what type of attribute is used?

Answer: Continuous numeric attribute (Ratio), as height can take any value in a range and has an absolute zero.



14- If the correlation coefficient between bread and rice = 1.5, what does it mean?

Answer: A correlation coefficient of 1.5 is invalid (must be between -1 and 1). Assuming a typo (e.g., 0.5), it indicates a moderate positive relationship between bread and rice sales.

15- If the covariance between bread and rice = 1, what does it mean?

Answer: Covariance = 1 indicates a positive relationship; when bread sales exceed their mean, rice sales tend to exceed theirs. Covariance doesn't indicate strength due to unit dependency.

16- If the information gain for attributes Age and Income are 0.24 and 0.024, which one would you select as splitting attribute?

Answer: Select Age, as it has higher information gain ($0.24 > 0.024$), reducing entropy more effectively in decision trees.

17- If the lift measure for bread and rice = 0.5, what does it mean?

Answer: Lift = 0.5 indicates a negative correlation; the likelihood of buying bread and rice together is 50% less than if they were independent.

18- If the lift measure for bread and rice = 1, what does it mean?

Answer: Lift = 1 indicates independence; buying bread does not affect the likelihood of buying rice.

19- If the lift measure for bread and rice = 1.5, what does it mean?

Answer: Lift = 1.5 indicates a positive correlation; buying bread increases the likelihood of buying rice by 50% compared to independence.

20- If the mean equals the median, what does it indicate about the data?

Answer: It indicates a symmetric distribution (e.g., normal distribution), with no significant skewness.

21- If the mean is larger than the median, what does it indicate about the data?

Answer: It indicates a positively skewed (right-skewed) distribution, with large outliers pulling the mean upward.

22- If the mean is less than the median, what does it indicate about the data?

Answer: It indicates a negatively skewed (left-skewed) distribution, with small outliers pulling the mean downward.

23- If I have 100 values in the dataset and add 5.0 to all values, how does it affect the median?

Answer: The median increases by 5.0, as adding a constant shifts all values equally without changing their order



24- If I have 100 values in the dataset and add 5.0 to all values, how does it affect the median?

Answer: مكرر من سؤال 23

25- List cluster analysis methods.

Answer:

- 1- Partitioning (e.g., K-Means).
- 2- Hierarchical (e.g., Agglomerative, Divisive).
- 3- Density-Based (e.g., DBSCAN).
- 4- Grid-Based (e.g., STING).
- 5- Model-Based (e.g., Gaussian Mixture Models).

26- List the major preprocessing tasks that improve data quality.

Answer:

- 1-Data Cleaning: Handling missing values, noise, errors.
- 2-Data Integration: Merging data from multiple sources.
- 3-Data Transformation: Normalization, aggregation.
- 4- Data Reduction: Dimensionality reduction, sampling.
- 5- Discretization: Converting continuous to discrete data.

27. List the steps of knowledge discovery.

Answer:

Data Cleaning // Data Integration // Data Selection // Data Transformation // Data Mining
// Pattern Evaluation // Knowledge Presentation.

28- List transformation strategies.

Answer:

Smoothing (e.g., binning, regression)
Aggregation (e.g., summarizing data).
Normalization (e.g., min-max, z-score)
Discretization (e.g., binning continuous values)
Attribute Construction (e.g., creating new features).



29- List types of outliers. Give an example for each type.

Answer:

Global Outlier: Deviates from the entire dataset.

Example: Grade of 10 in a 60-90 range.

Contextual Outlier: Deviates in a specific context.

Example: 30°C in winter in a cold city.

Collective Outlier: A group deviating together.

Example: Series of delayed orders in one day.

30- The confidence for the association rule {bread} → {milk, diapers} = 0.95. What does this value mean?

Answer: Confidence = 0.95 means 95% of transactions containing bread also contain milk and diapers.

31- The support for the association rule {bread} → {milk, diapers} = 0.95. What does this value mean?

Answer: Support = 0.95 means 95% of all transactions contain bread, milk, and diapers together.

32- What are rules conflicts? How can we resolve them?

Answer: Rules conflicts occur when multiple rules predict different outcomes for the same case (e.g., $\{A\} \rightarrow \{B\}$ vs. $\{A\} \rightarrow \{\neg B\}$)

- Select rule with highest confidence/support
- Prune weak rules using lift/conviction.
- Combine rules via weighted voting

33- What are data smoothing techniques?

Answer:

- Binning: Grouping values into bins, replacing with bin mean/boundaries.
- Regression: Fitting a model to smooth data.
- Clustering: Replacing outliers with cluster representatives.

34- What are data reduction strategies?

Answer:

- Dimensionality Reduction (e.g., PCA).
- Numerosity Reduction (e.g., sampling, regression).
- Data Compression (e.g., wavelet transforms).
- Discretization (e.g., binning).



35- What are the main advantages and disadvantages of decision tree classification algorithms?

Answer:

Advantages: Easy to interpret, handles numeric/categorical data, fast training/prediction.

Disadvantages: Unstable to small changes, prone to overfitting, biased toward attributes with many values.

36- What are the terminating conditions in constructing a decision tree?

Answer:

1-All records in a node belong to the same class.

2-No attributes left for splitting.

3-Node size below minimum threshold.

4-Information gain below threshold

5- Maximum tree depth reached.

37. What are the classifiers that are considered easily interpretable?

Answer:

1-Decision Trees: Clear if-then rules.

2-Naïve Bayes: Simple probabilistic rules.

3-Rule-Based Classifiers: Logical rules.

38- What are the clustering algorithms that can discover clusters with arbitrary shapes?

Answer:

DBSCAN // OPTICS // HDBSCAN

39. What is the data mining task that must be used to discover fraudulent credit card usage?

Answer: Outlier Analysis, to detect transactions deviating from normal patterns (e.g., large amounts or unusual locations)

40. What is overfitting? Explain one method to prevent it in classification trees.

Answer:

Overfitting occurs when a classifier learns noise/details in training data, reducing generalization.

Method: Pruning removes unnecessary branches (pre-pruning stops early; post-pruning removes after construction).



41. What is the Apriori property?

Answer: The Apriori property states that if an itemset is infrequent, all its supersets are also infrequent, reducing candidate itemsets in the Apriori algorithm

42. What is Bootstrap Sampling?

Answer: Bootstrap Sampling is random sampling with replacement, creating multiple datasets of the same size for model evaluation or ensemble methods (e.g., Bagging).

43. What is the difference between noise and outlier?

Answer: Noise is random errors/variations in data (e.g., measurement errors). Outliers are significantly different points, potentially meaningful (e.g., fraud). Noise is removed; outliers are analyzed.

44. What is the difference between a symmetric binary attribute and an asymmetric binary attribute?

Answer:

Symmetric Binary: Both values (0, 1) are equally important (e.g., gender: male=1, female=0).

Asymmetric Binary: One value (e.g., 1) is more significant (e.g., disease test: positive=1, negative=0).

45. What is the five-number summary of data? How is it represented visually?

Answer: Includes: Minimum, Q1 (25th percentile), Median, Q3 (75th percentile), Maximum.

Visual Representation: Boxplot (box from Q1 to Q3, median line, whiskers to min/max, outliers as points).

46. What is Majority Voting? When do we use it?

Answer: Majority Voting combines predictions from multiple classifiers in ensemble methods (e.g., Random Forests), selecting the class with the most votes. Used to improve classification accuracy with diverse models.

47. What does the association rule Computer → Webcam (60%, 100%) mean?

Answer: Support = 60%: 60% of transactions contain both Computer and Webcam.

Confidence = 100%: All transactions with Computer also contain Webcam.

48. What is the mode of data? What do Bimodal and Trimodal mean?

Answer:

Mode: the most frequent value.

Bimodal: Two values tie for highest frequency.

Trimodal: Three values tie for highest frequency.



49. What is the problem that occurs when we compute the mean? How can we solve it?

Answer:

Problem: Mean is sensitive to outliers, skewing results.

Solutions: Use median (robust to outliers), analyze outliers with boxplots, or use weighted mean

50. What is the problem that occurs when we use a global constant to fill missing values?

Answer:

Problem: Introduces bias, as the constant (e.g., “Unknown”) doesn’t reflect data variability.

Solutions: Use attribute mean/median, predict values with decision trees, or manual filling.

51. What is a redundant attribute? How can we detect it?

Answer: A redundant attribute provides unnecessary or derivable information (e.g., “age” and “birth date”). Detection: Correlation analysis (high correlation), PCA, or feature selection (e.g., Information Gain).

52. What are the types of data that can be mined?

Answer:

- Database-Oriented: Relational databases, data warehouses, transactional data.
- Advanced Data: Data streams, time-series, spatial, multimedia, text, web.
- Others: Social networks, graphs, legacy database

53. What are the types of patterns that can be mined?

Answer:

- Generalization.
- Association/Correlation Analysis.
- Classification.
- Cluster Analysis.
- Outlier Analysis.

54. When are two objects q and m density-connected?

Answer: Objects q and m are density-connected in DBSCAN if there’s a chain of core points connecting them, where each core point has MinPts neighbors within ϵ distance.

55. When is an object p density-reachable from an object q ?

Answer: Object p is density-reachable from q in DBSCAN if there’s a chain of core points from q to p , each within ϵ distance and having MinPts neighbors.



PRACTICAL BANK

Choose the correct answer:

1.	DataFrame contains data about the fourth-year students, How do you know the number of students ?			
	a) df.shape	b) df.shape[0]	c) df.shape[1]	d) len(df)
2.	DataFrame contains employees' details such as name, department, ..., and salary. How do you know the average salaries of all employees?			
	a) df.describe()	b) df.info()	c) df.sales.mean()	d) None
3.	To show the <u>relationship</u> between age, sales and reveal the distribution trends , you should use _____.			
	a) Line Chart	b) Pie Chart	c) Column Chart	d) Scatter plot
4.	Python library of useful tools for frequent patterns and associations			
	a) Pandas	b) NumPy	c) mlxtend	d) matplotlib
5.	Series in Pandas is _____-dimensional array.			
	a) one	b) two	c) three	d) None
6.	Which of the following is used for testing for <u>membership in the list of column names</u> ?			
	a) elif	b) in	c) out	d) None
7.	print(9//2); the output will be _____.			
	a) 4.0	b) 4.5	c) 4	d) Error
8.	_____ used to compute the nth percentile of the given data (array elements) along the specified axis.			
	a) percentile	b) sum	c) Div	d) Std
9.	How do we install NumPy in the system?			
	a) install numpy	b) pip install python numpy	c) pip install numpy	d) pip install numpy python
10.	_____ is an inbuilt function in Python programming language that returns the square root of any number			
	a) print	b) pass	c) sqrt	d) None
11.	_____ <u>limits the number of rounds</u> in k-means algorithm			
	a) iteration	b) tolerance	c) min	d) None
12.	Confusion matrix implementation in python needs _____ package.			
	a) NumPy	b) SymPy	c) pandas	d) sklearn
13.	_____ is a graphical representation of data where the individual values contained in a matrix of colors .			
	a) Histogram	b) Boxplot	c) Heatmap	d) Density Plot
14.	_____ shows the distribution of a numerical variable and really close to a histogram .			
	a) Scatter Plot	b) Density Plot	c) Violin Plot	d) Boxplot
15.	If you define List x = [1, 2, 3, 4, 5]; x[:3] returns _____. (x[:n] returns list from first element to n-1)			
	a) [1,2,3]	b) 1,2,3	c) [3,4,5]	d) Error



16.	The keyword _____ ends a function and give value of previous expression .			
	a) return	b) back	c) retrieve	d) break
17.	df.describe() command used to _____.			
	a) define data	b) summarize data	c) find the header	d) find no. of rows
18.	What data type is the object below? L = [1, 23, 'hello', 1]			
	a) Array	b) Tuple	c) Dictionary	d) List
19.	_____ which can be used to calculate the standard deviation			
	a) std	b) dts	c) tds	d) None
20.	How can we <u>convert the NumPy array to the list</u> in python?			
	a) list(array)	b) list.array	c) list	d) None
21.	import pandas as pan; Which of the following is the correct way to use DataFrame?			
	a) pan.DataFrame	b) DataFrame	c) pandas.DataFrame	d) None
22.	To return a single variable (username), you write any of the following?			
	a) df.username	b) df["username"]	c) df.loc[:, "username"]	d) All of the above
23.	Pandas is used to create _____.			
	a) a GUI programming	b) a database	c) a High-level array	d) All of the above
24.	Which of the following object you get after reading CSV file?			
	a) DataFrame	b) Character Vector	c) Panel	d) All of the above
25.	Which of the following keyword is used to access the numpy module in python?			
	a) access	b) import	c) fetch	d) from
26.	_____ is a standardized way of displaying the distribution of data based on a five-number summary ("minimum", first quartile (Q1), median, third quartile (Q3), "maximum")			
	a) Boxplot	b) Boxprint	c) Boxview	d) None
27.	The _____ can be a helpful tool in determining the <u>strength of the relationship</u> between two variables in linear regression			
	a) Euclidean distance	b) Least square	c) Extrapolation	d) Scatter plot
28.	If you define List x = [1, 2, 3, 4, 5]; x[-1] returns _____. (x[-1] returns last element which is 5, x[-2] returns second last which is 4)			
	a) 4	b) 2	c) 5	d) Error
29.	Function starts with _____ keyword.			
	a) def	b) function	c) import	d) try
30.	The if and else statements end with _____.			
	a) colon (:)	b) semicolon (;)	c) comma (,)	d) set brackets ({})
31.	df.shape command used to find _____.			
	a) No. of rows	b) No. of columns	c) No. of rows and No. of column	d) type of data file



32.	!pip list command used to _____.							
	a)	define list	b)	list of installed packages	c)	information about installed package	d)	print list
33.	Which of the following is not valid to import the numpy module?							
	a)	import numpy as np	b)	import numpy as n	c)	import numpy as p	d)	None
34.	_____ returns an arranged list of the <i>specified inerrable</i> object.							
	a)	sub	b)	sum	c)	div	d)	sorted
35.	_____ used for adding elements to List; It is used to add elements to the last position of List.							
	a)	append	b)	des	c)	asc	d)	None
36.	_____ package is one of the package that needed to be imported in implementing decision trees in python.							
	a)	NumPy	b)	SymPy	c)	pandas	d)	SciPy
37.	Assume you use DataFrame to describe the objects, How do you retrieve the used properties' names?							
	a)	df.describe()	b)	df.index	c)	df.columns	d)	df.info()
38.	Returns the items and itemsets with at least 60% support.							
	a)	apriori(df, min_support=60%)						
	b)	apriori(df, min_support=0.6)						
	c)	apriori(df, support=60%)						
	d)	apriori(df, support=0.6)						
39.	We can analyze the data in pandas with _____.							
	a)	Series	b)	DataFrame	c)	Both	d)	None
40.	_____ function can be used to <u>calculate mean</u> of a given list of numbers.							
	a)	sum	b)	min	c)	average	d)	max
41.	The method used to remove null values is _____.							
	a)	.dropna()	b)	.fillna()	c)	Both	d)	None
42.	The _____ method replaces the null values with a specified value.							
	a)	.dropna()	b)	.fillna()	c)	Both	d)	None
43.	What is the input to the predict() method that used in classification?							
	a)	The features of the data point	b)	The label of the data point	c)	Both the features and the label	d)	The confusion matrix
44.	From sklearn.metrics, you can import _____.							
	a)	Confusion matrix	b)	Accuracy score	c)	Classification report	d)	All of the above
45.	The scatter() function in <u>matplotlib</u> library is used to draw a <u>scatter plot</u> .							
	a)	TRUE	b)	FALSE	c)		d)	
46.	To calculate Sum of Squared Error (SSE) in Kmeans we use _____ attribute							
	a)	inertia_	b)	cluster_centers_	c)	n_iter_	d)	labels_



47.	Which of the following methods is used to display something like text, variables?			
	a) print()	b) input()	c) type()	d) int()
48.	Which statement is correct to import all modules from the package?			
	a) from package import all	b) from package import *	c) from package include all	d) from package include *
49.	To get row names of dataframe 'df', you can use _____.			
	a) df.index	b) df.rows	c) df.axes	d) df.axis
50.	When might the Apriori algorithm be faster than the FP-growth algorithm for association rule mining?			
	a) When the dataset is very large and dense			
	b) When the support threshold for frequent itemsets is very low			
	c) When the maximum itemset size is very large			
	d) When the dataset is very sparse			
51.	We can set _____ to convert these integer values in the itemsets column into the respective item names.			
	a) use_colnames=True	b) use_colname=True	c) colnames=True	d) None
52.	The _____ method is used to get the length of the itemsets.			
	a) apply(lambda x: len(x))	b) len(x)	c) len(lambda x)	d) None
53.	Pandas is a high-level data manipulation tool that is built on the NumPy package.			
	a) TRUE	b) FALSE	c)	d)
54.	Data Frames are incredibly powerful as they allow you to store and manipulate tabular data in rows of observations and columns.			
	a) TRUE	b) FALSE	c)	d)
55.	To compute silhouette_score in Kmeans, we need to import _____ module first.			
	a) sklearn.metrics	b) sklearn.neighbors	c) sklearn.naive_bayes	d) sklearn.cluster
56.	Silhouette coefficient values range between -1 and 1			
	a) TRUE	b) FALSE	c)	d)
57.	The input to the KNeighborsClassifier is the _____.			
	a) No. of neighbors	b) Accuracy score	c) No. of clusters	d) No. of features in the dataset
58.	From sklearn, you can import			
	a) A decision tree classifier	b) A naïve bayes classifier	c) A K-neighbors classifier	d) All of the above
59.	DataFrame.dropna(how='any'); is used to drop row or column if all values are null (any value is null)			
	a) TRUE	b) FALSE	c)	d)



60.	<pre>a=pd.read_table("C:\\Users\\raya\\Desktop\\dataset1withmissing.txt") f=a.fillna(method='bfill') print(f)</pre> The output of the code is _____. (bfill: backward fill, all null values replaced with NEXT values) ffill: forward fill, all null values replaced with PREVIOUS values)							
	a)	DataFrame object where all null values replaced with next values	b)	DataFrame object where all null values replaced with previous values	c)	Both	d)	None
61.	Which of the following libraries in Python is commonly used for association rule mining?							
	a)	NumPy	b)	Matplotlib	c)	Scikit-learn	d)	Apriori
62.	Which of the following is the wrong statement?							
	a)	When a data frame object is created, all the columns are sorted automatically	b)	To select a variable by column position, you can use data.iloc function	c)	del statement of data frame can be used to delete rows	d)	To get the structure of the data frame, you can use df.info()
63.	Which of the following statements is used to create an empty set in Python?							
	a)	()	b)	[]	c)	{}	d)	Set()
64.	What are the two main types of functions in Python?							
	a)	System function	b)	Custom function	c)	Built-in function & User-defined function	d)	User function
65.	For the following code line, kmeans_model.n_iter_, The Output is _____							
	a)	Coordinates of cluster centers	b)	Sum of Squared Error (SSE)	c)	Number of iterations	d)	Label of each point
66.	The n_neighbors parameter is used with KMeans function to specify number of clusters. (n_clusters)							
	a)	true	b)	false	c)		d)	
67.	The predict() method is used to <u>train</u> the classification model. (fit() method)							
	a)	true	b)	false	c)		d)	
68.	From SKlearn.neighbors, you can import _____							
	a)	Decision tree classifier	b)	Naive Bayes classifier	c)	Gaussian classifier	d)	None of the mentioned classifiers
69.	_____ Method is used to count the number of missing values for each variable in dataframe							
	a)	.isnull()	b)	.isnull().sum()	c)	.isnull().sum().sum()	d)	None of them
70.	_____ Method is used to count the number of missing values in the whole dataset							
	a)	.isnull()	b)	.isnull().sum()	c)	.isnull().sum().sum()	d)	None of them
71.	.isnull() method returns a dataframe object where all the values are replaced with boolean values							
	a)	true	b)	false	c)		d)	



72.	.isnull() method returns boolean values with value _____ for null values, and otherwise _____							
	a)	True, False	b)	False, True	c)	True, True	d)	False, False
73.	Which keyword is used for function in Python language?							
	a)	function	b)	def	c)	Fun	d)	Define
74.	Which of the following character is used to give single-line comments in Python?							
	a)	//	b)	#	c)	!	d)	/*
75.	Which of the following statements is used to print coordinates of cluster centers in K-means?							
	a)	kmeans_model .n_iter_	b)	kmeans_model .inertia_	c)	kmeans_model .labels_	d)	kmeans_model .cluster_centers_
76.	Which of the following statements is used to give label for each point in dataset in K-means?							
	a)	kmeans_model .inertia_	b)	kmeans_model .labels_	c)	r_centers_	d)	kmeans_model .n_iter_
77.	The fit() method is used to test the classification model. (train)							
	a)	true	b)	false	c)		d)	
78.	How many inputs for the fit() method used to build any classification model?							
	a)	It takes no input	b)	One input	c)	Two inputs	d)	More than two inputs
79.	Select the command to get the dimension of dataframe 'df'.							
	a)	df.dim	b)	df.dimension	c)	df.shape	d)	df.len
80.	What is the name of the data transformation technique that converts categorical variables into binary variables for association rule mining?							
	a)	OneHot encoding	b)	Label encoding	c)	Standardization	d)	Normalization

